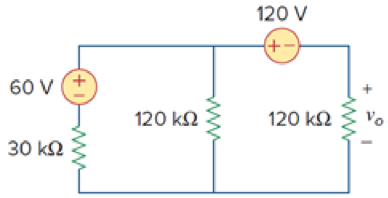
Chapter 3, Problem 5P

Problem

Obtain *vo* in the circuit of Fig. 3.54.

Figure 3.54:



Step-by-step solution

Step 1 of 4

Refer to Figure 3.54 in the text book.

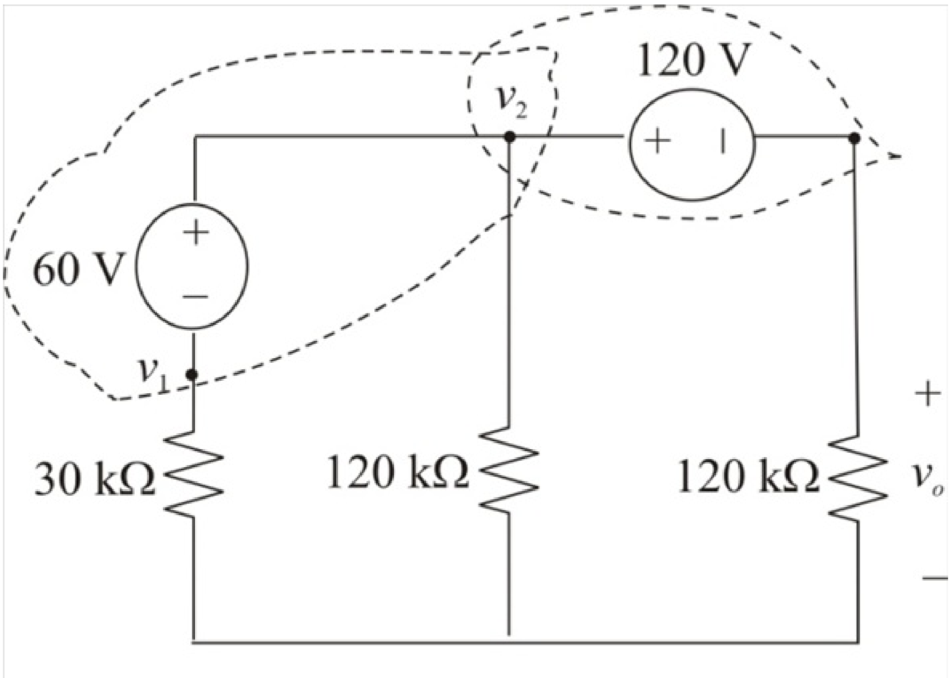


Figure 1

Comments (3)

Anonymous

Why were these nodes chosen?

Anonymous

^^^

Anonymous

Why are the nodes placed there?

Step 2 of 4

Consider the relation between the voltages v_o and v_2.

$$v_2 - v_o = 120$$
$$v_2 = 120 + v_o$$

Therefore, the expression for the voltage v_2 is, $120 + v_o$.

Step 3 of 4

Consider the relation between the voltages v_1 and v_2.

$$v_2 - v_1 = 60$$
$$v_1 = v_2 - 60$$

Substitute $120 + v_o$ for v_2 in the equation.

$$\begin{aligned} v_1 &= 120 + v_o - 60 \\ &= 60 + v_o \end{aligned}$$

Therefore, the expression for the voltage v_1 is, $60 + v_o$.

Step 4 of 4

Apply super node analysis at all indicated nodes.

$$\frac{v_2}{120\text{ k}} + \frac{v_o}{120\text{ k}} + \frac{v_1}{30\text{ k}} = 0$$
$$\frac{v_2 + v_o + 4v_1}{120\text{ k}} = 0$$
$$v_2 + v_o + 4v_1 = 0$$

Substitute $60 + v_o$ for v_1 and $120 + v_o$ for v_2 in the equation.

$$120 + v_o + v_o + 4(60 + v_o) = 0$$
$$120 + v_o + v_o + 240 + 4v_o = 0$$
$$360 + 6v_o = 0$$
$$6v_o = -360$$
$$\begin{aligned} v_o &= -\frac{360}{6} \\ &= -60\text{ V} \end{aligned}$$

Therefore, the value of the voltage, v_o is $\boxed{-60\text{ V}}$.

Comments (3)

Anonymous

where did the 120k go?

Anonymous

the second equation in step 4

Anonymous

nvm i got it